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HERIOT WATT UNIVERSITY

SCHOOL OF SOCIAL SCIENCES

Edinburgh Business School

C27IE

INTRODUCTORY ECONOMICS

August Examination and Assessment Diet 2024

Duration: Two hours

Section A: Multiple choice questions

There are 20 questions in this section. Please answer all questions at the start of your answer paper. Each question is worth 1.5 marks.

1) The issues that economists study stem from the central economic problem of scarcity. Which of the following is true?

   a. Scarcity will eventually disappear with the development of new technology and higher levels of production.
   b. Scarcity is an issue for producers, but not for consumers.
   c. The opportunity cost principle is important in production, but not for consumers.
   d. Whenever there is scarcity, we need to choose which goods and services we desire most.

2) When increasing amounts of a variable factor are used with a given amount of a fixed factor, there will come a point when each extra unit of the variable factor will produce less additional output than the previous unit. We can then say that use of the factor is subject to:

   a. Diminishing factor intensity
   b. Diminishing marginal cost
   c. Diminishing marginal returns
   d. Diseconomies of scale

3) At the current market price of 120, there is an excess supply of running shoes. We expect that:

   a. The market price will increase, the quantity supplied will increase, and the quantity demanded will decrease.
   b. The market price will decrease, the quantity supplied will increase, and the quantity demanded will decrease.
   c. The market price will decrease, the quantity supplied will decrease, and the quantity demanded will increase.
   d. The market price will increase, the quantity supplied will decrease, and the quantity demanded will increase.

4) Shell produces both petrol and diesel fuels, which are used in different types of engine, so that there is very little possibility of substitution between them (in the short-run). We might expect to find a value of the cross-price elasticity of demand for these products which is:

a. Greater than 1
b. Between 0 and 1
c. Between -1 and 0
d. Less than -1

5) Widespread storms in Brazil have led to a shortage of coffee beans. This can be illustrated by:

a. a leftward shift in the demand curve and a fall in the equilibrium price.
b. a rightward shift in the supply curve and a fall in the equilibrium price.
c. a leftward shift in the supply curve and an increase in the equilibrium price.
d. a rightward shift in the demand curve and an increase in the equilibrium price.

6) Which one of the following would make collusion in an oligopoly more likely?

a. The firms have similar cost structures and production methods.
b. The industry has four firms, who operate separate supply chains
c. Each firm is secretive about costs and production methods
d. The industry has no significant barriers to entry

7) KTM is the main train operator in Malaysia, and so has substantial monopoly power. As a result, we expect that:

a. It will run fewer services than if it faced competition from other businesses.
b. It will not make economic profits in the long-run.
c. It will set prices equal to marginal costs to maximise its profits.
d. It will set lower train fares than if it faced competition from other businesses

8) If we are thinking about production in the long run, which of the following is true?

a. The firm can hire more workers but cannot build a new factory.
b. It is possible that the firm will face decreasing returns to scale.
c. With constant returns to scale, a 1% increase in inputs will lead to a larger percentage increase in output.
d. There will be diminishing marginal returns to some factor.

9) We would expect the demand for a specific brand of coffee beans to be more price elastic than the demand for coffee beans as a product because:

a. Not all consumers are able to grind coffee beans.
b. The market for the brand of coffee beans is much smaller than the market for all coffee beans.
c. Consumers develop a taste for their preferred brand of coffee beans.
d. There is a much higher degree of substitutability among brands of coffee beans than for coffee beans as a product.

10) We define a public good as having the properties of non-excludability and non-rivalry in consumption. Which of the following services comes closest to meeting those requirements

a) Street lighting
b) A golf club.
c) An inter-city coach service
d) A public park

11) After the government imposes a tariff on imports of steel, which group can we be certain will be worse off?

a. Domestic firms which produce steel.
b. The government.
c. Workers employed in the steel industry.
d. Firms which use imported steel in production.

12) Which one of these is a supply side tool for reducing the natural rate of unemployment?

a) Lowering the central bank's interest rate for overnight lending to commercial banks
b) Abolishing licencing qualifications for skilled workers
c) Decreasing taxes on earned income and employment
d) Increasing the government's capital expenditure

13) Which one of these is an example of an expansionary fiscal policy?

a) Reducing the central bank's interest rates
b) Financing increased government expenditure by increasing the money

supply
c) Increasing the income which can be earned without paying income taxes
d) Introducing subsidies for the export of manufactured goods

14) Which of the following is not a basis for calculating the GDP measure of national income?
a) The balance on the capital and current accounts with the rest of the world
b) The expenditure on all final goods and services produced within the economy.
c) The incomes received by households from the provision of factors of production.
d) The value added in the production of goods and services.

15) The real interest rate is

a) the interest rate corrected for deflation
b) the interest rate corrected for unemployment
c) the interest rate corrected for investment
d) the interest rate corrected for inflation

16) Given that the velocity of circulation of money in an economy is 12, real GDP is 1,000bn and the money supply is 95bn. According to the Quantity Theory of Money, the price level will be:

a) 0.86
b) 1
c) 1.14
d) 1.28

17) In which one of these situations might real GDP decrease and the price level remain constant?

a) The aggregate demand and aggregate supply curves both shift right by the same amount.
b) The aggregate demand and aggregate supply curves both shift left by the same amount.
c) The aggregate demand curve does not move and the aggregate supply curve shifts left.
d) The aggregate demand curve shifts left and the aggregate supply curve does not move.

18) Government policy leads to interest rates falling, and both consumption and investment expenditure increasing. With which one of these policies are

these effects consistent?

a. A monetary expansion.
b. A fiscal expansion.
c. A fiscal contraction.
d. A monetary contraction.

19) If taxation is 20% of income, there is no saving, and one quarter of disposable income (income after tax has been paid) is spent on imports, then the marginal propensity to spend on domestic goods is:

a. 0.75
b. 0.6
c. 0.55
d. 0.4

20) Which of the following might be expected to shift the aggregate supply curve to the right?

a. The removal of free nursery education for young children.
b. A long period during which interest rates are high.
c. A reduction in government subsidies to firms.
d. The creation of a national apprenticeship scheme.

Section B: Short answer questions

There are four questions in this section, each worth 10 marks. You should answer all four questions.

1) Many countries experienced substantial increases in food prices during 2022.

a) Define the concept of the price elasticity of demand, indicating how it is measured.

(1 marks)

b) If the price of rice has increased from 1.2 per kg to 1.8 per kg, and the demand of a typical household has fallen from 5.5kg per month to 4.5kg

per month, calculate the household price elasticity of demand for rise using the mid-point method.

(3 marks)

c) State whether the demand for rice, based on the calculation in part b), is elastic or inelastic and explain why this might be the case.

(1 marks)

d) Data shows that the overall demand for rice has decreased by 10 per cent over the last five years and this coincides with a period during which real incomes have risen by 15 per cent. Calculate the income elasticity of demand for rice and use the calculation to state whether it is an inferior good or a normal good.

(3 marks)

(Question 1 Total: 10 marks)

2. Crystal Clear Audio plans to produce a range of video cameras. It has estimated how the number of cameras which it could produce in a year will change with the number of workers which it hires.

a) Define the average physical product and the marginal physical product. Complete the table below.

[4 marks]

Workers	0	20	40	60	80	100	120
Output (thousands)	0	60	140	210	270	320	360
APP	-						
MPP							

b) Sketch a diagram showing the average physical product and the marginal physical product. In your sketch, show clearly where the average physical product and the marginal physical product are equal.

[4 marks]

c) Assume that Crystal Clear Audio pays its workers 30,000. Given that wages are the only variable costs, complete the table below, showing the average variable cost of output.

[2 marks]

Output (thousands)	0	60	140	210	270	320	360
TVC (thousands)							
AVC	-						

3) Governments can use monetary policy to manage aggregate demand

a. Explain why we treat an increase in interest rates as a form of contractionary monetary policy. Briefly outline how this type of policy can help to reduce the rate of inflation.

(4 marks)

b. The UK, and other major economies, have recently suffered from 'stagflation', a combination of high inflation and low economic growth. Explain why the central bank may be torn between adopting contractionary or expansionary monetary policy in these circumstances.

(3 marks)

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c. Explain what would happen to the value of the Euro if the European Central Bank were to set interest at a higher level than in other countries.

(3 marks)

(Question 3 Total: 10 marks)

4) You are given the following information on an open economy:

Consumer expenditure = 100 + 0.8Y
Investment expenditure = 600
Government expenditure = 1000
Exports = 500
Imports = 0.3Y
There are no taxes

a) Calculate the equilibrium level of income.

(4 marks)

b) Suppose that the government believes that the full employment level of income Y^* = 4,800. It proposes to reach that level by encouraging

investment.

i. Explain briefly why the increase in national income after an increase in investment is larger than the increase in investment.

(2 marks)

ii. The multiplier, $m = \Delta Y/\Delta I$, is a measure of the extent to which national income increases relative to the increase in investment. State the relationship between the multiplier and the marginal propensity to withdraw income from the circular flow, and then calculate the value of the investment multiplier in this example.

(3 marks)

iii. By how much should investment increase if the economy is to reach its full employment level of national income, $Y^* = 4{,}800$?

(1 mark)

(Question 4 Total: 10 marks)

Section C – Discussion questions

There are five questions in this section, each of which is worth fifteen marks. You should answer any two questions.

1) Outline the elements of a PEST analysis for a global airline company, such as Emirates Airlines, which is based in Dubai.

2) When the price of oil increases, the prices of inputs in food production also tend to increase rapidly. An example of this was in 2022, when fertiliser prices increased substantially. In the same year, wholesale grain prices also increased substantially. Discuss how it was possible for farmers to achieve record profits in 2022.

3) Applying the structure-conduct-performance approach, what might we conclude about the likely structure of an industry, such as electric power generation in which there are substantial economies of scale?

4) For many years, economists have used the term ‘business cycle’ to refer to the succession of periods of economic expansion, followed by recession, over approximately 7 – 9 years. An important objective of government

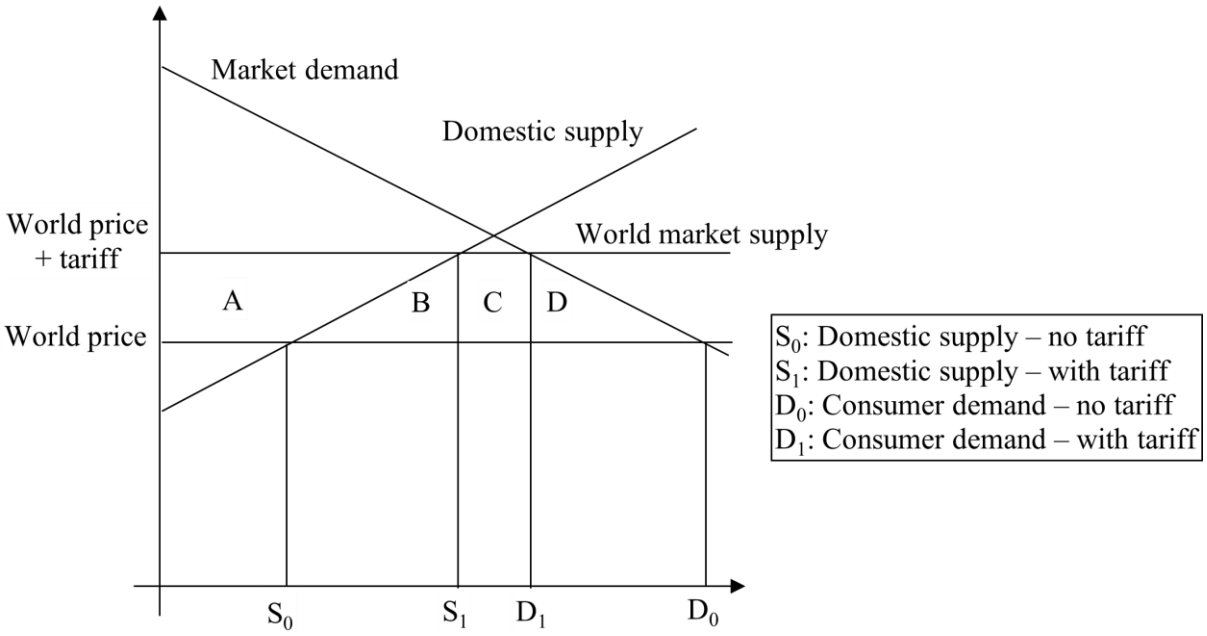
economic policy is to manage these fluctuations.

a. Briefly outline the links between the business cycle and the credit cycle in banking. [6 marks]

b. How might governments design their policies so that measures to reduce the fluctuations in the business cycle either occur automatically or else follow rules? [9 marks]

[Total: 15 marks]

5) Outline the economic argument represented in this diagram, to explain how a tariff benefits certain groups in society. Explain briefly why governments may be willing to use tariffs as a policy tool.



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